

SET 6

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the questions given.

1. (a) Evaluate the following limits, if exist.

(i)
$$\lim_{x \rightarrow 2} \frac{x-2}{x^4-16}$$

[3 marks]

(ii)
$$\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$$

[3 marks]

- (b) Consider a function
- $f(x) = \frac{1}{2-\sqrt{x}}$
- . Find
- $\lim_{x \rightarrow \infty} f(x)$
- and state the equation of horizontal asymptotes for
- $f(x)$
- .

[3 marks]

2. (a) Consider the parametric equations,
- $x = t^2$
- , and
- $y = t^3 - 3t$
- .

Evaluate $\frac{dy}{dx}$ when $t = \sqrt{3}$.

[5 marks]

- (b) Given
- $y = 3^{2x-5}$
- . Find
- $\frac{d^2y}{dx^2}$
- when
- $x = 3$
- . Give the answer in the form of logarithm.

[4 marks]

3. The function
- $f(x) = x^3 - 6x^2 + 9x - 3$
- is defined on the interval
- $[0, 5]$
- . Find the critical points of
- $f(x)$
- on this interval and determine whether the critical points are local minimum or maximum.

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** the questions given.

1. Solve for
- p
- and
- q
- where
- $p \neq q$
- , such that
- $\frac{(p+qi)}{3i} = (3 + \sqrt{-16} - i^3)$
- .

[5 marks]

2. (a) Solve the equation
- $\log_2(x) - \log_4(3x+4) = 0$
- .

[4 marks]

- (b) Find the solution set for the inequality
- $\frac{-4-x}{x-3} \geq x+4$
- ,
- $x \neq 3$
- .

[7 marks]

3. Given
- $f(x) = \frac{x^2+1}{5}$
- ,
- $x \geq 0$
- .

- (a) Determine $f^{-1}(x)$. Hence, if $f(g(x)) = \frac{1}{5}(e^{2(3x-1)} + 1)$, show that $g(x) = e^{3x-1}$.

[5 marks]

- (b) Evaluate $g(f(2))$ correct to three decimal places.

[2 marks]

4. Given $f(x) = e^x$ and $g(x) = |x-3|$.

- (a) Show that $(f \circ g)(x) = \begin{cases} e^{x-3}, & x \geq 3 \\ e^{-(x-3)}, & x < 3 \end{cases}$.

[3 marks]

- (b) Determine $(f \circ g)^{-1}(x)$, for $x \geq 3$.

[4 marks]

- (c) Find the function $h(x)$ for $x > \frac{1}{3}$, given that $(h \circ f)(x) = \frac{2e^x}{1-3e^x}$.

Hence, show that $h(x)$ is a one to one function.

[6 marks]

5. (a) Given $f(x) = \frac{2|x|}{x} + 5x$. Compute $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$.

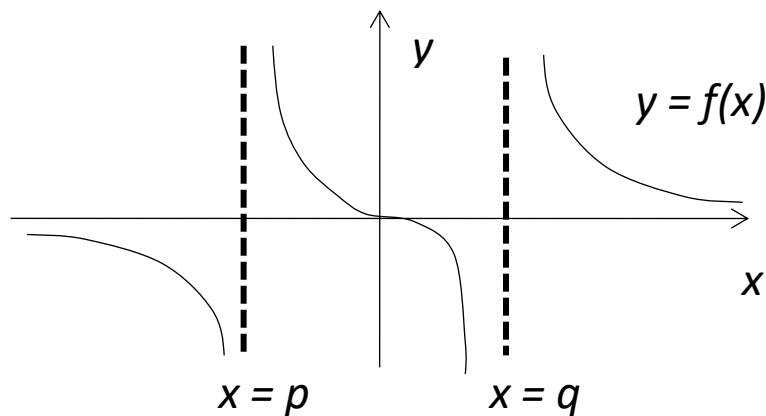
[4 marks]

- (b) The continuous g is defined by $g(x) = \begin{cases} \sqrt{5-x}, & x < a, \\ 3x-1, & x \geq a \end{cases}$.

Find the value of a .

[6 marks]

- (c) In the graph below, let $f(x) = \frac{2x}{x^2-4}$.



- (i) From the graph, $x = p$ and $x = q$ are the vertical asymptotes. Find the values p and q .

- (ii) Determine the horizontal asymptote.

[4 marks]

6. By writing $\tan x$ in terms of $\sin x$ and $\cos x$, show that $\frac{d}{dx}(\tan x) = \sec^2 x$. [3 marks]
- (a) If $y = \tan x$, find $\frac{d^2y}{dx^2}$ in terms of y . Hence, determine the range of value of x such that $\frac{d^2y}{dx^2} > 0$ for $0 < x < \pi$. [7 marks]
- (b) If $y = \tan(x + y)$, find $\frac{dy}{dx}$ in terms of x and y . Hence show that $\frac{dy}{dx} = -\operatorname{cosec}^2 2\alpha$ when $x = y = \alpha$. [6 marks]
7. (a) Use the derivative to find the maximum area of a rectangle that can be inscribed in a semicircle of radius 10 cm. [5 marks]
- (b) The side of a cube of length x cm is increasing at the rate of 2.5cm/second. How fast is the rate of volume of the cube increasing when the side is 8 cm long? [4 marks]

END OF QUESTION PAPER

Final Answer

SECTION A

1. (a) (i) $\frac{1}{32}$ (ii) -1
- (b) $\lim_{x \rightarrow \infty} f(x) = 0$, $y = 0$ is H.A
2. (a) $\sqrt{3}$
- (b) $12(\ln 3)^2$
3. Critical points $(1,1)$ and $(3,-3)$
Maximum point : $(1,1)$; Minimum point : $(3,-3)$

SECTION B

1. $p = -15, q = 9$
2. (a) $x = 4$
- (b) $\{x : x \leq -4 \cup 2 \leq x < 3\}$

3. (a) $\sqrt{5x-1}$, show
(b) 7.389
4. (a) Show
(b) $\ln x + 3$
(c) $\frac{2x}{1-3x}$, $h(x)$ is one to one function
5. (a) 2, -2
(b) $a = 1$
(c) (i) $p = -2$, $q = 2$ (ii) $y = 0$
6. (a) $2(1 + y^2)y$, $0 < x < \frac{\pi}{2}$
(b) $\frac{\sec^2(x + y)}{1 - \sec^2(x + y)}$
7. (a) 100 cm^2
(b) $480 \text{ cm}^3 / \text{second}$